Programming project report CSC223

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for Professor Sally Schafner

https://github.com/ApexArcadian/analyze-search-algorithms.git

DATA COLLECTED:

size(N)	RBS	IBS	SS
5000	7.02	3.91	385.76
50000	10.01	3.99	3045.59
100000	20.43	6.61	8113.75
150000	11.9	6.01	7177.59
1000000	16.17	9.78	58737.18

Which search method was fastest/slowest and why? (Relate to Big-O.)

IBS is by far the fastest, with sequential being the slowest.
Seq is O(N) so at larger data sizes it struggles.
Both binary search algorithms have O(log n) time. Even at N=1000000 it takes ~10-20 microseconds because log base 2 of 1000000 is about 20.

Which binary search variant was faster? Why, even with the same Big-O?

Iterative is ~1.5-2x faster even though they have the same time complexity because recurisve methods must deal with function call overhead. Each level of recursion adds a new function call which is more memory intesive than an additional loop iteration.

Compare the execution times for N = 5000 with N = 1000000 and provide a reason for the differences.

It is immediately obvious that a larger sample size will increase run times for a sequential search if the target is not one of the first elements.

Why might sequential search find a target at a different location?

If a target is present at multiple indexes a sequential search might find it at an earlier point than other search algorithms.

What was the most challenging part of this project?

Organizing the data after analysis.

What did you learn?

I learned how to measure algorithm performance with pythons time module and perf_counter.